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GATE ECE · CONTROL SYSTEMS · CHAPTER-WISE PYQS

Root Locus Diagram

GATE ECE Control Systems · Previous Year Questions with Answer Key

23	21	1988 – 2025	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	8 questions
Marks 2	15 questions
Answer key	at the end

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Marks 1 8 questions

Q1. GATE ECE 2025

Open solution

Consider the unity-negative-feedback system shown in Figure (i) below, where gain $K \geq 0$. The root locus of this system is shown in Figure (ii) below. For what value(s) of K will the system in Figure (i) have a pole at $-1 + j1$?

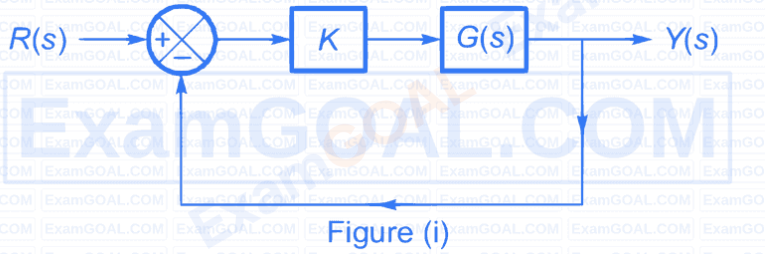


Figure (i)

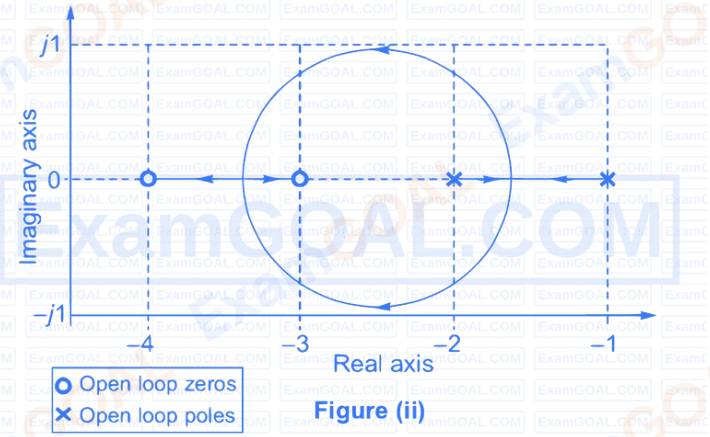


Figure (ii)

(A) $K = 5$

(B) $K = \frac{1}{5}$

(C) For no positive value of K

(D) For all positive values of K

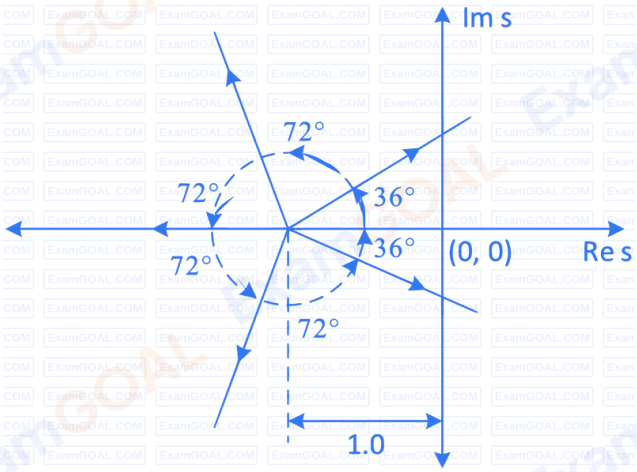
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Q2. GATE ECE 2022

Open solution

The root-locus plot of a closed-loop system with unity negative feedback and transfer function KG(s) in the forward path is shown in the figure. Note that K is varied from 0 to ∞.

Select the transfer function G(s) that results in the root-locus plot of the closed-loop system as shown in the figure.



(A) $G(s) = \frac{1}{(s+1)^5}$

(B) $G(s) = \frac{1}{s^5+1}$

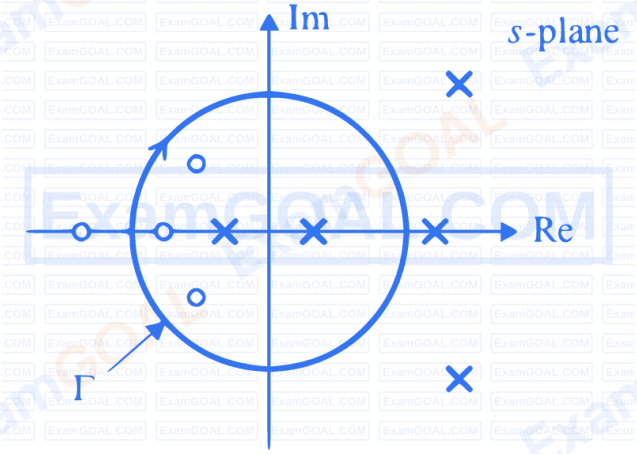
(C) $G(s) = \frac{s-1}{(s+1)^6}$

(D) $G(s) = \frac{s+1}{s^6+1}$

Q3. GATE ECE 2020

Open solution

The pole-zero map of a rational function $G(s)$ is shown below. When the closed contour Γ is mapped into $G(s)$-plane, then the mapping encircles



(A) the point $-1 + j0$ of the $G(s)$-plane once in clockwise direction.

(B) the origin of $G(s)$-plane once in counter - clockwise direction.

(C) the point $-1 + j0$ of the $G(s)$-plane once in counter - clockwise direction.

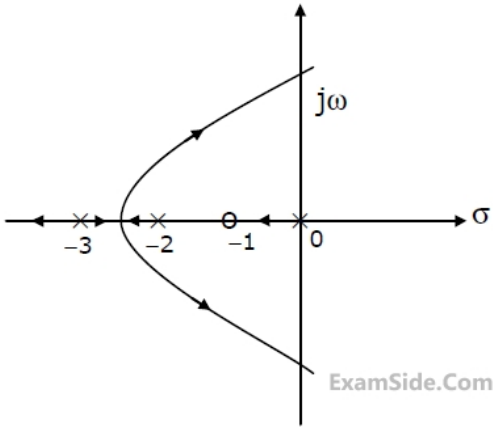
(D) the origin of $G(s)$-plane once in clockwise direction.

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Q4. GATE ECE 2011

The root locus plot for a system is given below. The open loop transfer function corresponding to this plot is given by

Open solution



(A) $G(s)H(s) = k\frac{s(s+1)}{(s+2)(s+3)}$

(B) $G(s)H(s) = k\frac{(s+1)}{s(s+2)(s+3)^2}$

(C) $G(s)H(s) = k\frac{1}{s(s-1)(s+2)(s+3)}$

(D) $G(s)H(s) = k\frac{(s+1)}{s(s+2)(s+3)}$

Q5. GATE ECE 2002

Which of the following points is NOT on the root locus of a system with the open loop transfer function

Open solution

$$G(s)H(s) = \frac{K}{s(s+1)(s+3)}$$

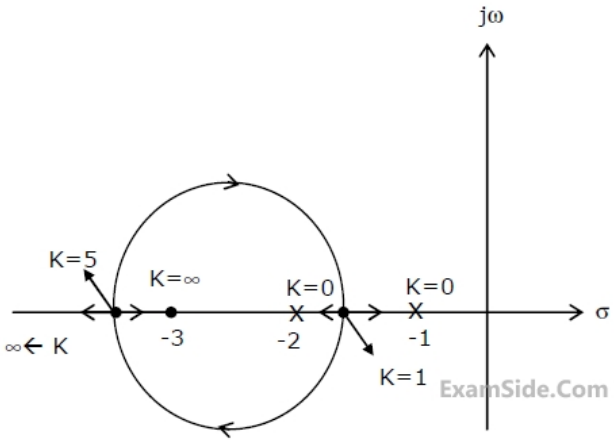
(A) $s = -j\sqrt{3}$ (B) $s = -1.5$ (C) $s = -3$ (D) $s = -\infty$

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Q6. GATE ECE 2001

The root-locus diagram for a closed loop feedback system is shown in Figure. The system is overdamped.

Open solution



(A) $only\ if\ 0 \leq K \leq 1$

(B) $only\ if\ 1 < K < 5$

(C) $only\ if\ K > 5$

(D) $if\ 0 \leq K < 1\ or\ K > 5$

Q7. GATE ECE 2001

Given the $G(s)H(s) = \frac{K}{s(s+1)(s+3)}$, the point of intersection of the asymptotes of the root loci with the real axis is

Open solution

(A) -4

(B) 1.33

(C) -1.33

(D) 4

Q8. GATE ECE 1994

If the open loop transfer function is a ratio of a numerator polynomial of degree 'm' and a denominator polynomial of degree 'n', then the integer (n-m) represents the number of

Open solution

(A) breakaway points

(B) unstable poles

(C) separate root loci

(D) asymptotes

Marks 2 15 questions

Q9. GATE ECE 2020

The characteristic equation of a system is

$s^3 + 3s^2 + (K+2)s + 3K = 0$

In the root locus plot for the given system, as K varies from 0 to ∞, the break-away or break-in point(s) lie within

Open solution

(A) $-\infty, -3$

(B) $-3, -2$

(C) $-1, 0$

(D) $-2, -1$

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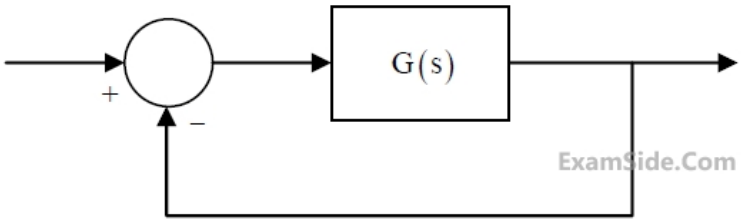
Q10. GATE ECE 2017 Set 1

Open solution

A linear time invariant (LTI) system with the transfer function

$$G(s) = \frac{K(s^2 + 2s + 2)}{s^2 - 3s + 2}$$

is connected in unity feedback configuration as shown in the figure.



For the closed loop system shown, the root locus for 0 < K < ∞ intersects the imaginary axis for K = 1.5. The closed loop system is stable for

(A) K>1.5
(B) 1 < K < 1.5
(C) 0 < K < 1
(D) no positive value of K

Q11. GATE ECE 2016 Set 1

Open solution

The open-loop transfer function of a unity-feedback control system is

$$G(s) = \frac{K}{s^2 + 5s + 5}$$

The value of K at the breakaway point of the feedback control system's root-locus plot is _____

Q12. GATE ECE 2016 Set 3

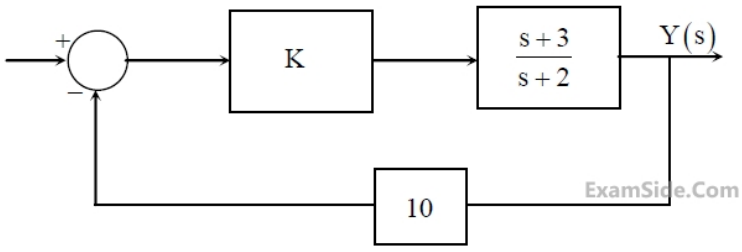
Open solution

The forward-path transfer function and the feedback-path transfer function of a single loop negative feedback control system are given as $G(s) = \frac{K(s+2)}{s^2+2s+2}$ and H(s) = 1 respectively. If the variable parameter K is real positive, then the location of the breakaway point on the root locus diagram of the system is_____.

Q13. GATE ECE 2015 Set 3

Open solution

For the system shown in the figure, s = −2.75 lies on the root locus if K is____.



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Q14. GATE ECE 2015 Set 1

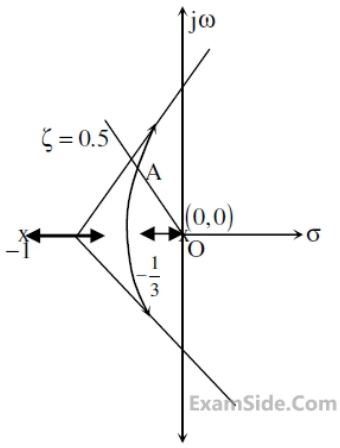
Open solution

The open-loop transfer function of a plant in a unity feedback configuration is given as $G(s) = \frac{K(s+4)}{(s+8)(s^2-9)}$. The value of the gain K(>0) for which -1 + j2 lies on the root locus is _____.

Q15. GATE ECE 2014 Set 4

Open solution

The characteristic equation of a unity negative feedback system 1 + KG(s) = 0. The open loop transfer function G(s) has one pole at 0 and two poles at -1. The root locus of the system for varying K is shown in the figure.

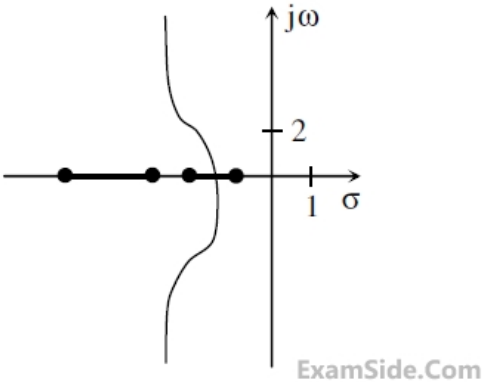


The constant damping ratio line, for ξ = 0.5 , intersects the root locus at point A. The distance from the origin to point A is given as 0.5. The value of K at point A is _____ .

Q16. GATE ECE 2014 Set 3

Open solution

In the root locus plot shown in the figure, the pole/zero marks and the arrows have been removed. Which one of the following transfer functions has this root locus?



(A) $\frac{s+1}{(s+2)(s+4)(s+7)}$

(B) $\frac{s+4}{(s+1)(s+2)(s+7)}$

(C) $\frac{s+7}{(s+1)(s+2)(s+4)}$

(D) $\frac{(s+1)(s+2)}{(s+4)(s+7)}$

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Q17. GATE ECE 2007

Open solution

A unity feedback control system has an open-loop transfer function

$$G(s) = \frac{K}{s(s^2 + 7s + 12)}$$

The gain K for which s = −1 + j1 will lie on the root locus of this system is:

(A) 4 (B) 5.5 (C) 6.5 (D) 10

Q18. GATE ECE 2005

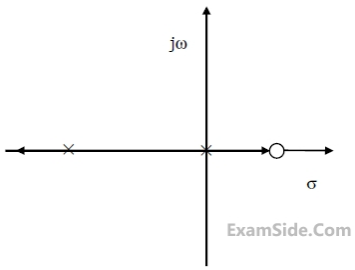
Open solution

A unity feedback system is given as,

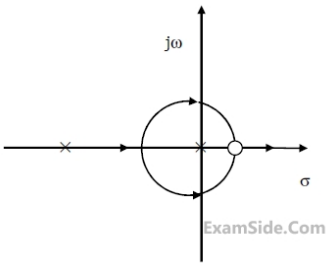
$$G(s) = \frac{K(1 - s)}{s(s + 3)}$$

indicate the correct root locus diagram.

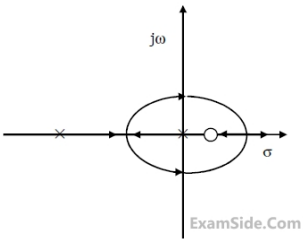
(A)



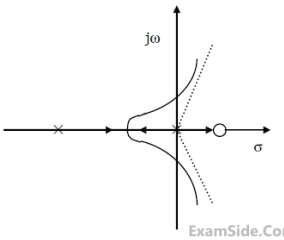
(B)



(C)



(D)



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Q19. GATE ECE 2003

Open solution

The root locus of the system

$$G(s)H(s) = \frac{K}{s(s+2)(s+3)}$$

has the break-away point located at

(A) (-0.5,0) (B) (-2.548,0) (C) (-4,0) (D) (-0.784,0)

Q20. GATE ECE 1999

Open solution

Consider the points $s_1 = -3 + j4$ and $s_2 = -3 - j2$ in the s-plane. Then, for a system with the open loop transfer function

$$G(s)H(s) = \frac{K}{(s+1)^4}$$

(A) s_1 is on the root locus, but not s_2

(B) s_2 is on the root locus, but not s_1

(C) both s_1 and s_2 are on the root locus

(D) neither s_1 nor s_2 is on the root locus

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Q21. GATE ECE 1992

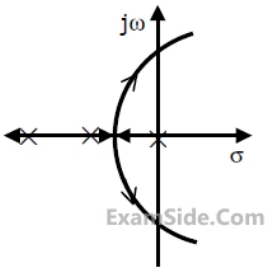
Open solution

Given a unity feedback system with open loop transfer function,

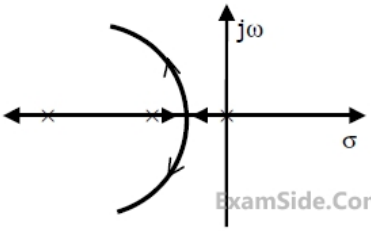
$$G(s) = \frac{K}{s(s+1)(s+2)}$$

The root locus plot of the system is of the form.

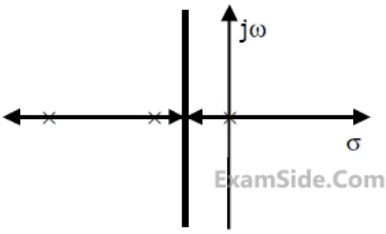
(A)



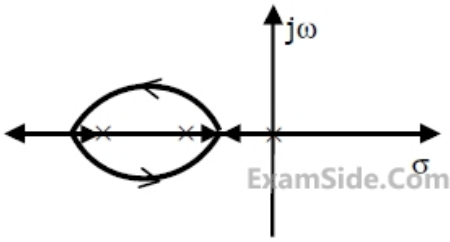
(B)



(C)



(D)



Q22. GATE ECE 1991

Open solution

The characteristic equation of a feedback control system is given by

$s^3 + 5s^2 + (K + 6)s + K = 0$

Where K > 0 is a scalar variable parameter. In the root loci diagram of the system the asymptotes of the root locus for large values of K meet at a point in the s-plane whose coordinates are

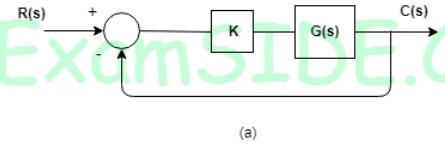
(A) (-3,0) (B) (-2,0) (C) (-1,0) (D) (2,0)

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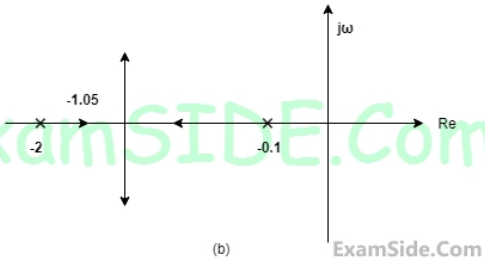
Q23. GATE ECE 1988

Open solution

Consider a closed-loop system shown in fig. (a) below. The root locus for it is shown in fig. (b). The closed-loop transfer function for the system is



(a)



(b)

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(A) $\frac{K}{1+(0.5s+1)(10s+1)}$

(B) $\frac{K}{(s+2)(s+0.1)}$

(C) $\frac{K}{1+K(0.5s+1)(10s+1)}$

(D) $\frac{K}{K+0.2(0.5s+1)(10s+1)}$

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Answer Key

Marks 1

1. **C**	2. **A**	3. **D**	4. **B**	5. **B**
6. **D**	7. **C**	8. **D**		

Marks 2

9. **C**	10. **A**	11. **1.25**	12. **-3.414**	13. **0.3**
14. **25.54**	15. **0.375**	16. **B**	17. **D**	18. **C**
19. **D**	20. **B**	21. **A**	22. **B**	23. **D**

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